

Meta-analysis of the effect of β -glucan intake on blood cholesterol and glucose levels.

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OBJECTIVE: A meta-analysis was performed on epidemiologic studies to assess the relation between β -glucan consumption from oats and from barley on blood cholesterol level, triglyceride/triacylglycerol (TGL/TAG) level, and blood glucose level (BGL) in humans. In addition, the effect of β -glucan on total cholesterol (TC) and BGL was translated into an empirical dose-response model. METHODS: Thirty research articles that evaluated the effect of different exposure levels of β -glucan on blood cholesterol and BGL were analyzed, yielding 126 clinical studies. RESULTS: There was a significant inverse relation in TC (-0.60 mmol/L, 95% confidence interval [CI] -0.85 to -0.34), low-density lipoprotein (-0.66 mmol/L, 95% CI -0.96 to -0.36), and TGL/TAG (-0.04 mmol/L, 95% CI -0.15 to 0.07) after consumption of β -glucan. In contrast, an increase in high-density lipoprotein cholesterol was noted (0.03 mmol/L, 95% CI -0.06 to 0.13) with the random-effect model. The analysis showed a significant change in BGL (-2.58 mmol/L, 95% CI -3.22 to -1.84) with high heterogeneity between ($I(2) = 97\%$) and across ($\tau(2) = 5.88$) the studies. The fixed-effect model showed a significant change in TC, low-density lipoprotein, and BGL, whereas it showed no significant changes in high-density lipoprotein and TGL/TAG. The dose-response model showed that a 3-g/d dose of oat or barley β -glucan was sufficient to decrease TC. CONCLUSION: Consumption of 3 g/d of oat or barley β -glucan is sufficient to decrease blood cholesterol, whereas the effect on BGL is still inconclusive, with high heterogeneity, and requires further clinical research studies with longer intervention periods. Copyright © 2011 Elsevier Inc. All rights reserved.